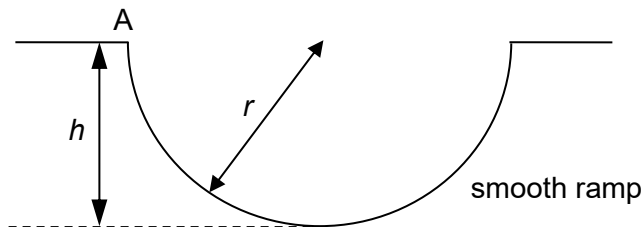


- 11 A skateboarder rests on top of a smooth ramp of height h and radius of curvature r . He moves off from his position of rest at A and slides down the smooth ramp.



The mass of the skateboarder is m .

What is the magnitude of the contact force R exerted by the ramp on the skateboarder at the lowest point of the ramp?

- A mg
- B $2mg$
- C $3mg$
- D $4mg$

12 A satellite of mass 50 kg moves from a point where the gravitational potential due to the